MODEL LAB · 2026-10-28

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)	11	17,800	2.92%	0.5998
served prior	priors compared	held-out plate appearances	held-out HR rate	best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-13, scored on the 17,800 that came after (actual HR rate 2.92%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5813	1.62×	0.02826	0.13096	0.00454
KNN comps Prior is the pooled rate of the nearest comparables.	0.5998	1.64×	0.02821	0.12987	0.00605
SVR Support-vector regression on the same feature matrix.	0.5925	1.52×	0.02826	0.13049	0.00693
Random forest Forest regression; a learned similarity metric.	0.5951	1.50×	0.02825	0.13018	0.00678
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5998	1.64×	0.02821	0.12987	0.00605
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5998	1.64×	0.02821	0.12987	0.00605
Linear regression Ordinary least squares on the same features.	0.5964	1.58×	0.02825	0.13026	0.00487
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5953	1.60×	0.02830	0.13055	0.00609
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5955	1.56×	0.02824	0.13016	0.00503
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5945	1.45×	0.02829	0.13055	0.00733

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **260** knn **260** svr **260** rf **260** core **260** ensemble **260** linear **260** logistic **260** nn **260** xgb **260**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (1,620 PA, actual HR rate 2.35%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.6456	1.58×	0.02291	0.11126	0.01790
KNN comps Prior is the pooled rate of the nearest comparables.	0.6062	1.58×	0.02288	0.10980	0.01007
SVR Support-vector regression on the same feature matrix.	0.5783	1.58×	0.02323	0.11613	0.02309
Random forest Forest regression; a learned similarity metric.	0.6000	1.32×	0.02320	0.11319	0.02260
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6062	1.58×	0.02288	0.10980	0.01007
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6062	1.58×	0.02288	0.10980	0.01007
Linear regression Ordinary least squares on the same features.	0.6112	1.32×	0.02322	0.11383	0.02025
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6015	1.32×	0.02354	0.11442	0.02384
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6056	1.58×	0.02300	0.11128	0.01557
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5934	1.05×	0.02361	0.11640	0.02762

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jackson Chourio 16.8%, William Contreras 15.5%, Jake Bauers 15.4%, Junior Caminero 14.5%, Christian Yelich 12.9%, Garrett Mitchell 11.9%
Flat league	5/6	Jackson Chourio 17.1%, William Contreras 15.4%, Jake Bauers 14.3%, Junior Caminero 13.9%, Christian Yelich 12.9%, Yandy Díaz 11.9%
KNN comps	5/6	Jackson Chourio 17.4%, William Contreras 15.6%, Jake Bauers 15.6%, Junior Caminero 14.3%, Christian Yelich 13.2%, Jonathan Aranda 11.9%
SVR	5/6	Jackson Chourio 17.7%, Junior Caminero 15.6%, William Contreras 15.5%, Jake Bauers 15.3%, Christian Yelich 12.6%, Yandy Díaz 12.1%
Random forest	5/6	Jackson Chourio 17.6%, Jake Bauers 15.6%, William Contreras 15.4%, Junior Caminero 14.9%, Christian Yelich 12.9%, Yandy Díaz 11.9%
Core (KNN+SVR)	5/6	Jackson Chourio 17.4%, William Contreras 15.6%, Jake Bauers 15.6%, Junior Caminero 14.3%, Christian Yelich 13.2%, Jonathan Aranda 11.9%
Forest blend	5/6	Jackson Chourio 17.4%, William Contreras 15.6%, Jake Bauers 15.6%, Junior Caminero 14.3%, Christian Yelich 13.2%, Jonathan Aranda 11.9%
Linear regression	5/6	Jackson Chourio 17.6%, William Contreras 15.4%, Jake Bauers 15.3%, Junior Caminero 14.5%, Christian Yelich 13.1%, Jonathan Aranda 11.8%
Logistic regression	5/6	Jackson Chourio 17.6%, William Contreras 15.5%, Jake Bauers 15.3%, Junior Caminero 14.9%, Christian Yelich 12.6%, Yandy Díaz 11.7%
Neural net (torch)	5/6	Jackson Chourio 17.6%, William Contreras 15.6%, Jake Bauers 15.5%, Junior Caminero 14.9%, Christian Yelich 12.7%, Yandy Díaz 11.7%
XGBoost	5/6	Jackson Chourio 17.6%, Jake Bauers 15.6%, William Contreras 15.4%, Junior Caminero 15.1%, Christian Yelich 12.8%, Yandy Díaz 12.0%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Jackson Chourio** MIL slate	**16.8%**	17.1%	17.4%	17.7%	17.6%	17.4%	17.4%	17.6%	17.6%	
2	**William Contreras** MIL	**15.5%**	15.4%	15.6%	15.5%	15.4%	15.6%	15.6%	15.4%	15.5%	
3	**Jake Bauers** MIL	**15.4%**	14.3%	15.6%	15.3%	15.6%	15.6%	15.6%	15.3%	15.3%	
4	**Junior Caminero** TB slate	**14.5%**	13.9%	14.3%	15.6%	14.9%	14.3%	14.3%	14.5%	14.9%	
5	**Christian Yelich** MIL	**12.9%**	12.9%	13.2%	12.6%	12.9%	13.2%	13.2%	13.1%	12.6%	
6	**Garrett Mitchell** MIL	**11.9%**	11.5%	11.7%	11.2%	11.5%	11.7%	11.7%	11.4%	11.4%	
7	**Jonathan Aranda** TB	**11.8%**	11.2%	11.9%	11.5%	11.5%	11.9%	11.9%	11.8%	11.6%	
8	**Yandy Díaz** TB	**11.6%**	11.9%	11.7%	12.1%	11.9%	11.7%	11.7%	11.8%	11.7%	
9	**Brice Turang** MIL	**11.6%**	11.7%	11.3%	11.5%	11.2%	11.3%	11.3%	10.8%	10.8%	
10	**Victor Mesa Jr.** TB	**9.5%**	9.2%	9.5%	9.8%	9.9%	9.5%	9.5%	9.7%	9.6%	
11	**Cooper Pratt** MIL	**8.1%**	9.5%	7.9%	7.4%	7.1%	7.9%	7.9%	6.8%	7.0%	
12	**David Hamilton** MIL	**7.4%**	8.7%	7.2%	7.0%	6.6%	7.2%	7.2%	7.2%	7.1%	
13	**Jonny DeLuca** TB	**6.9%**	7.9%	7.0%	6.8%	6.6%	7.0%	7.0%	7.0%	7.3%	
14	**Liam Hicks** TB	**6.5%**	7.4%	6.5%	7.2%	7.0%	6.5%	6.5%	6.9%	6.7%	
15	**Sal Frelick** MIL	**6.4%**	8.0%	6.7%	6.1%	5.9%	6.7%	6.7%	6.4%	6.5%	
16	**Richie Palacios** TB	**5.4%**	6.0%	5.3%	5.1%	4.6%	5.3%	5.3%	5.0%	5.1%	
17	**Taylor Walls** TB	**3.7%**	4.9%	3.6%	2.7%	2.7%	3.6%	3.6%	3.4%	3.5%	
18	**Chandler Simpson** TB	**2.5%**	5.4%	3.3%	1.5%	1.7%	3.3%	3.3%	1.4%	2.8%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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